Exploratory Visualization of Fraud Risk Patterns in Bank Transaction Data

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Abstract

Bank fraud is an important problem for banks because fraudulent transactions are often mixed with many normal transactions. In this project, we will use a synthetic bank transaction fraud dataset from Kaggle to explore which transaction features may be related to higher fraud risk. We will use exploratory data analysis to compare fraudulent and non-fraudulent transactions through visualizations. The analysis will focus on transaction amount, transaction time, night transactions, international transactions, failed attempts, payment methods, merchant categories, device types, countries, and fraud types. We hope to find useful patterns that show what kinds of transactions are more common in fraud cases within this dataset.

Problem

Financial fraud can cause losses for banks and customers, and it can also reduce customer trust in digital banking services. As digital transactions become increasingly common, financial institutions must process large volumes of transactions every day, making it difficult to identify suspicious activities hidden among millions of legitimate transactions.

A key challenge is that fraudulent transactions rarely follow a single, consistent pattern. Instead, fraud may be associated with complex interactions among transaction, customer, and behavioral characteristics. These relationships are often difficult to recognize using raw data tables or simple summary statistics alone. This project aims to use exploratory data analysis and visualization to answer the question: **What transaction characteristics are visually associated with higher fraud risk in bank transaction data?**

Intended Contribution

The main contribution of this project is to use data visualization to explore fraud risk patterns in bank transaction data. Instead of focusing only on prediction, this project will focus on understanding and explaining the patterns in the dataset. We will compare fraudulent and non-fraudulent transactions and identify which transaction features are more common in fraud cases.

The analysis will include transaction-related features such as amount, time, night transaction status, international transaction status, failed attempts, and time since the last transaction. It will also compare categorical features such as payment method, merchant category, device type, country, city, and fraud type. The final contribution will be a set of visualizations that make fraud risk patterns easier to understand in this synthetic dataset.

Methodology

This project uses the Bank Transaction Fraud Detection dataset from Kaggle, consisting of 1,000,000 synthetic bank transaction records and fraud labels. The analysis will be conducted in R using **RStudio** and **Quarto**, with **tidyverse, dplyr, ggplot2**, and **knitr** supporting data preparation, visualization, and reporting. Because the dataset is synthetic, the findings will be interpreted as patterns observed within the dataset rather than direct conclusions about real-world banking systems.

Data Preparation: The first stage of the analysis focuses on understanding the structure and quality of the dataset. Variable types, missing values, summary statistics, and the overall fraud distribution will be examined to identify potential data quality issues and establish a baseline understanding of the dataset. This stage will also include the identification of outliers and unusual observations that may influence subsequent analyses.

Exploratory Visualization: The second stage focuses on exploring the distributions of individual variables through univariate visualizations. Histograms, bar charts, boxplots, and violin plots will be used to examine transaction amounts, customer characteristics, account balances, and fraud-related variables. These visualizations will help identify dominant patterns, variability, and potential anomalies within the data.

Fraud Pattern Analysis: The final stage investigates relationships between transaction characteristics and fraud occurrence. Bivariate and multivariate visualizations will be used to compare fraudulent and legitimate transactions across variables such as transaction behavior, merchant category, payment method, device type, geographic location, and fraud type. The objective is to identify characteristics that are visually associated with higher fraud risk and to develop a comprehensive visual profile of potentially fraudulent transactions.

If time permits, a simple classification model such as logistic regression or random forest may be developed to evaluate whether the patterns identified during exploratory analysis are useful for fraud prediction. Model performance may be assessed using precision, recall, F1-score, and confusion matrices. However, the primary focus of the project remains exploratory data analysis and visualization.

Expected Results

We expect to find visible differences between fraudulent and non-fraudulent transactions. For example, fraud may appear more often in international transactions, night transactions, transactions with more failed attempts, or some specific merchant categories and payment methods. Transaction amount, device type, country, and city may also show different patterns between fraud and non-fraud cases.

We also expect that different fraud types may have different profiles. Some fraud types may be related to higher transaction amounts, while others may be more common in certain merchant categories, payment methods, or device types. The final results will include visualizations that show these patterns clearly and help answer the main research question. Since the dataset is synthetic, we will only explain the results within this dataset and will not claim that they directly represent real bank fraud behavior.

Team Responsibilities

Team Member	Responsibilities
Ngoc Khanh Vy Huynh	Write the Problem and Methodology sections; describe the dataset, tools, libraries, and analysis steps.
Keyao Li	Write the Abstract, Intended Contribution, and Expected Results sections; help organize the research question and expected findings.
Both members	Discuss the topic, check the final proposal, revise the writing, and prepare the final submission together.

Key References

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